Research Proposal

Title:
Recognition of Handwritten Bangla Upazila–District Pair Names Using Deep Ensemble Learning

1. Abstract

Automated recognition of handwritten Bangla administrative pair names (upazila–district) is an underexplored but essential task for postal automation, form digitization, and e-governance in Bangladesh. This research proposes to develop a complete pipeline: (i) collect a new handwritten dataset covering all 496 upazila–district pairs from a diverse set of writers, (ii) design an unsupervised segmentation method to extract text lines from multi-column handwritten pages, and (iii) build a recognition system using ensemble learning that combines multiple deep neural networks (e.g., CNNs, Vision Transformers, or hybrid architectures) to achieve high accuracy. The effectiveness of different ensemble fusion strategies (hard voting, soft voting, weighted voting, and stacking) will be systematically compared. The expected outcome is a robust, publicly available benchmark and a recognition model that can support real-world applications.

2. Problem Statement

Handwritten Bangla text recognition lags far behind Latin scripts due to limited datasets, complex character shapes, and connected components. Specifically, no dedicated system exists for recognizing **upazila–district pair names** – a crucial unit in Bangladeshi postal addresses. Without such a system, manual sorting remains slow and error-prone, hindering digital transformation in postal and government services.

3. Research Objectives

1. **Create the first publicly available dataset** of handwritten Bangla upazila–district pair names covering all 496 administrative units, collected from a diverse set of writers.
2. **Develop an unsupervised segmentation pipeline** to reliably extract individual text regions from scanned handwritten forms (which may contain multiple columns, lines, and overlapping text).
3. **Implement and compare multiple deep learning base models** (CNNs, Vision Transformers, CRNNs, etc.) for classifying the extracted handwritten pair names.
4. **Design an ensemble learning framework** that combines multiple base models to improve accuracy beyond any single model.
5. **Evaluate different ensemble fusion strategies** (majority voting, soft voting, weighted voting, stacking) and identify the most effective approach for this complex script.

4. Methodology

4.1 Data Collection and Annotation

- Collect handwritten samples from 300+ volunteers across different districts.
- Each writer fills pre-designed forms containing all 496 upazila–district pair names.
- Scan forms, segment manually (initial version), and annotate with class labels.

4.2 Segmentation Approach

- Design an unsupervised pipeline using MSER (Maximally Stable Extremal Regions), Canny edge detection, column analysis, row grouping, and contour merging to separate overlapping or touching text regions.

4.3 Recognition Models (To Be Explored)

- **Convolutional Neural Networks** (e.g., ResNet, DenseNet, ConvNeXt)
- **Vision Transformers** (e.g., ViT, DeiT)
- **Recurrent/Attention models** (e.g., CRNN, Transformer-based OCR)

4.4 Ensemble Learning

- Train several diverse base models (different architectures, preprocessing, or weight initializations).
- Combine their predictions using:
    - Hard voting (majority rule)
    - Soft voting (average of probabilities)
    - Weighted soft voting (learned or heuristic per-model weights)
    - Stacking (meta-learner on top of base models)

4.5 Evaluation Metrics

- Accuracy, precision, recall, F1-score (macro and weighted).
- Confusion matrix analysis to identify problematic classes.

5. Expected Outcomes

- A new publicly available dataset: **BNHW UDPNR** (Bangla Handwritten Upazila–District Pair Name Recognition).
- An unsupervised segmentation pipeline that reduces manual intervention.

- Performance comparison of multiple deep learning architectures and ensemble methods.
- A final ensemble model achieving **target accuracy > 90%** on the test set.
- Recommendations on the best ensemble strategy for low-resource handwritten Bangla recognition.

6. Timeline (6 Months)

Month	Activities
1	Data collection form design, writer recruitment, and sample collection
2	Scanning, annotation, dataset cleaning, baseline segmentation
3	Implement base models (CNNs, ViT) and train on the dataset
4	Develop ensemble methods (voting, stacking) and tune hyperparameters
5	Evaluate, compare, and select best model; write results
6	Final thesis writing and submission

8. Potential Risks and Mitigation

Risk	Mitigation
Low writer participation	Collaborate with local educational institutions and community centers
Class imbalance (rare upazilas)	Oversampling, data augmentation, or weighted loss functions
Poor segmentation	Use both unsupervised and semi-supervised methods; plan for limited manual correction
Model overfitting	Regularization, cross-validation, and early stopping

9. Conclusion

This research will fill a critical gap in Bangla handwritten text recognition by focusing on practically important upazila–district pair names. The proposed dataset, segmentation pipeline, and ensemble learning framework will provide a foundation for postal automation, digital governance, and future research on low-resource script OCR.